

Explore the Power of the Bench Calculator

This second article on the mathematical journey through open source takes you through the basics of the bench calculator.



Faced with the limitations of the shell command *expr* and other shell constructs, let's set out to explore the powerful command *bc*, which stands for *bench calculator*. This is not just a command or tool, but a complete language in itself. And its power lies in its arbitrary preciseness with not only integers, but with real numbers. If you're wondering what that means, its computation is usually not limited by the size of the integer or real number types, unlike in most programming languages. Thus, this is closer to our day-to-day use of mathematics, abstracting away the internal details of the computer's precision. So, let's get started with the first bits of math, and then move on to more involved aspects like variables, conditionals, and still later on, functions and recursion.

Basic operations

For integer-only math, you can invoke the bench calculator as '*bc*'. For full-fledged real number math, invoke it as '*bc -l*'. Once invoked, it will print a welcome message and then wait for you to type your math statements, before pressing '*Enter*' to get your answer. To quit *bc*, enter *Ctrl-D* on an empty line. All the basic arithmetic operations: addition (+), subtraction (-), multiplication (*), quotient (/), remainder (%), power (^), and brackets (()) are just there—with C-language-like precedence and associativity rules. An example with all of them in use is shown below:

```
$ bc
bc 1.06.95
```